Assignment 2 Product Architecture and Design for Manufacturing

Redesign and improve your manufacturing demonstrator in ManuLab

Purpose

Use product architecture to improve manufacturability. Analyse the current demonstrator, develop a modular architecture, implement selected changes in ManuLab, and verify the effect.

Main assignment question

How can the demonstrator's product architecture enable simpler, repeatable and scalable manufacturing while preserving required customer functions?

Learning outcomes

After the assignment, you should be able to:

- Visualise a product family in Customer View and Engineering View.
- Identify functions, modules, variants, interfaces and constraints.
- Explain effects on flow, assembly, quality, cost, flexibility and automation readiness.
- Apply Design for Manufacturing and Assembly to a physical demonstrator.
- Compare baseline and improved designs using observations and measurements.
- Discuss trade-offs between customer variety, complexity and manufacturing performance.

Expected result

Deliver a physically improved demonstrator, not only a concept. Show clearer module boundaries, better interfaces and a more repeatable manufacturing or assembly process.

Team and submission

Team assignment. Submit the report and supporting files in Blackboard by the stated deadline. Bring the improved demonstrator to the final ManuLab session.

Work process

Follow the four phases. Document key decisions, observations and test results. Use the Product Family Master Plan framework from the PPT.

Phase 1 Establish the baseline

1. Define the product boundary and main customer need.
2. Create an as-is Customer View: segments, values, selectable features and variety.
3. Create an as-is Engineering View: functions, components, modules, variants, attributes and constraints.
4. Map physical, mechanical, electrical and digital interfaces.
5. Classify the architecture as integral, modular or hybrid. Support the choice with evidence.
6. Observe manufacturing and assembly in ManuLab. Record access problems, adjustment, rework, waiting, variation, unclear interfaces and unnecessary handling.

Baseline measurements

Select at least four indicators, including one from each category below. Use three repeated trials where practical; otherwise explain the alternative evidence.

Category	Possible indicators
Time and flow	Assembly, setup, changeover or waiting time; process steps
Complexity	Parts, unique parts, interfaces, fastener types or assembly directions
Quality	Rework, adjustment, errors, fit problems, test failures or build variation
Flexibility	Variant change time, supported variants, scalable range or upgrade ease
Automation readiness	Self-location, gripping surfaces, access, orientation or repeatability

Phase 2 Develop the target architecture

1. Define target module boundaries and the variation assigned to each module.
2. Create target Customer and Engineering Views.
3. Specify critical interfaces: dimensions, connections, tolerances, information flow and ownership.
4. Show support for variety, flexibility, scalability, maintenance, replacement and upgrades.
5. Select at least two physical redesign actions for implementation.

Design for Manufacturing requirements

Apply principles that address the demonstrator's actual problems. Explain why they matter.

Design principle	Evidence expected in the redesign
Reduce unnecessary complexity	Fewer parts, unique parts, interfaces or operations
Standardise	Reusable modules; common components, dimensions, fasteners and interfaces
Control variety	Create customer variety through module choices, not full redesign

Design principle	Evidence expected in the redesign
Simplify assembly	Better access, fewer directions, self-location, simpler joining and less adjustment
Prevent errors	Poka-yoke, keyed interfaces, clear orientation and verification features
Design for automation	Stable geometry, repeatable positioning, good access and controlled variation
Design for service and upgrade	Replaceable modules, accessible connections and minimal disturbance

Phase 3 Implement the redesign in ManuLab

1. Update relevant CAD, drawings, bill of materials and work instructions.
2. Build and integrate at least two approved physical improvements.
3. Keep a change log: problem, decision, change and expected effect.
4. Photograph the baseline, implementation and final demonstrator from comparable viewpoints.
5. Verify that the redesigned product performs its required functions.

Phase 4 Verify and learn

1. Repeat the baseline measurements with the same method.
2. Compare before and after. Report observations and calculated changes.
3. Test critical interfaces: fit, function, repeatability and robustness.
4. Separate effects from architecture changes and local process changes.
5. Identify trade-offs and unintended effects, such as cost, weight, reduced flexibility or new constraints.
6. Recommend the next improvement for repeated industrial production.

Minimum evidence for an accepted physical improvement

- At least two design changes are implemented.
- Function is demonstrated before and after redesign.
- At least four manufacturing indicators are compared.
- Critical interfaces are defined and verified.
- Evidence links architecture decisions to manufacturing effects.

Required deliverables

No.	Deliverable	Required content
1	Team report	Maximum 10 pages, excluding cover, references and appendices.
2	PFMP Customer View	As-is and target: customer values, features, variety and configuration choices.
3	PFMP Engineering View	As-is and target: functions, components, modules, variants, attributes and constraints.
4	Architecture and interface map	Module boundaries, interface types, critical dimensions or tolerances, and ownership.
5	Physical demonstrator	At least two integrated DFM improvements, ready for demonstration.
6	Before-and-after evidence	Measurements, photos and comparison of at least four indicators.
7	Demonstration	5-minute live demonstration or 3-minute video: redesign, function and improvement.
8	Team contribution log	Short table of each member's responsibility and contribution.

Before-and-after results table

Include one summary table. Use measured values where possible; explain qualitative ratings.

Indicator	Baseline	Improved design	Change	Evidence
Assembly or process time				
Complexity indicator				
Quality or rework indicator				
Flexibility or changeover indicator				
Additional team-selected indicator				

Connection to the final exam delivery

For the final exam, update the target Customer and Engineering Views to match the built and tested demonstrator. Include final module and interface definitions, verified results and key lessons. Show changes made after testing; do not repeat the assignment unchanged.

Submission checklist

- Report in PDF format
- Editable PFMP and architecture figures
- Updated CAD, drawings or bill of materials where relevant
- Before-and-after measurement data and photographs
- Demonstration video if a live demonstration is not used